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Superseding
MIL-T-5494
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MILITARY SPECIFICATION

THERMOCOUPLES, CONTACT, AIRCRAFT ENGINE SPARK PLUG GASKET-TYPE

This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 Scope.- This specification covers iron-constantan gasket-type thermocouples.

1.2 Classification.- Thermocouples shall be one of the following sizes, as specified (see 6.2):

14 mm gasket
18 mm gasket

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standard, and drawings, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

LLL-B-631	Boxes; Fiber Corrugated (for Domestic Shipment)
LLL-B-636	Boxes, Fiber, Solid, (for Domestic Shipment)

Military

JAN-P-108	Packaging and Packing for Overseas Shipment - Boxes, Fiberboard (V-Board and W-Board), Exterior and Interior
MIL-P-6064	Packaging of Lightweight Aircraft Accessories

STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage

FSC 6685

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DRAWINGS

Air Force-Navy Aeronautical Standard Drawings

AN5540	Thermocouple - Aircraft Engine Gasket Type
AND10398	Metals - Definition of Dissimilar

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Qualification.- The thermocouples furnished under this specification shall be a product which has been tested and has passed the Qualification tests specified herein.

3.2 Materials.- Materials shall conform to applicable specifications as specified herein. Materials which are not covered by applicable specifications shall be of the best commercial quality and suitable for the purpose intended.

3.2.1 Metals.- All metals used in the construction of gasket-type thermocouples shall be of a corrosion-resistant type or shall be suitably protected to resist corrosion, except for the copper gasket which shall not be plated. The use of dissimilar metals, especially brass, copper, or steel in intimate metal-to-metal contact with aluminum or aluminum alloy, shall be avoided wherever practicable. Dissimilar metals are defined in Drawing AND10398.

3.3 Design and construction.- The thermocouple assembly shall be designed to produce an EMF equivalent to the temperature at the junction as specified in table I. The thermocouple shall be so constructed as to withstand the strains, jars, vibrations, and such other conditions as are incident to shipping, storage, installation, and service.

TABLE I

EMF output for iron-constantan thermocouple

Temperature °C	Electromotive force mv ±0.16
50	2.66
100	5.40
150	8.19
200	10.99
250	13.79
300	16.56

NOTE: Values are for reference junction temperature 0°C.

3.4 Size and weight.- The thermocouple shall conform to Drawing AN5540, and the weight shall not exceed 0.10 pound.

3.5 Wiring.-

3.5.1 Stranding.- Each iron wire and each constantan wire shall consist of at least 5 strands. The wire size of all strands shall be the same, the exact gage depending upon that required to bring the resistance to the value specified herein.

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3.5.2 Insulation.- The two conductors and outside insulation shall be suitably protected to resist deterioration from oil, cooling liquid, engine cleaning fluids, or heat, by a fiber-glass braid.

3.6 Temperature range.- The thermocouple assembly shall be constructed to function satisfactorily in the temperature range -55° to +300°C (-67° to +572°F).

3.7 Finish.- Protective coatings and finishes which will crack, chip, scale, or char during normal service life shall not be used. The individual wires shall not be tinned or otherwise finished.

3.8 Identification of product.-

3.8.1 Nameplate data.- The manufacturer's part number, manufacturer's name or trade-mark, and AN part number shall be permanently marked on the thermocouple. Marking may be accomplished by use of a sleeve over the lead wires, or by marking directly on the fiber-glass braid.

3.8.1.1 The code "NBS" shall be stamped on each thermocouple, as shown on Drawing AN5540, to indicate calibration wire in accordance with Bureau of Standards RP1080 curve (table I).

3.8.2 Use of AN or MIL designations.- AN or MIL designations shall not be applied to a product, except for Qualification test samples, nor referred to in correspondence, until notice of approval has been received from the activity responsible for qualification.

3.9 Performance.- The thermocouple shall satisfy the requirements specified in section 4, and shall not show material or structural failure when subjected to the following tests:

(a) Resistance	(4.4.1.2)	(d) Full-range EMF output	(4.3.2.1)
(b) EMF output	(4.4.2.1)	(e) Extreme temperature exposure	(4.3.2.2)
(c) Bending	(4.4.2.2)	(f) Vibration	(4.3.2.3)
		(g) Humidity	(4.3.2.4)

3.10 Workmanship.- All details of workmanship shall be in accordance with high-grade commercial practice for the manufacture of thermocouples.

4. QUALITY ASSURANCE PROVISIONS

4.1 Classification of tests.- The inspection and testing of thermocouples shall be classified as follows:

(a) Qualification tests: Qualification tests are those tests performed on samples submitted for approval as qualified products.

(b) Acceptance tests: Acceptance tests are those tests performed on individual lots which have been submitted for acceptance under contract.

4.2 Test conditions.-

4.2.1 Liquid bath.- The EMF output and Full-range EMF output tests as specified in 4.4.2.1 and 4.3.2.1, respectively, shall be conducted with the copper gasket of the thermocouple under test immersed in a bath of a suitable liquid material to give the desired range of 50° to 300°C (122° to 572°F).

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4.3 Qualification tests.-

4.3.1 Sampling instructions.- The Qualification test samples shall consist of four thermocouples of each manufacturer's part number upon which qualification is desired. Prior to submitting samples to the Government for approval, the manufacturer shall conduct Individual tests only. Samples shall be forwarded to the activity responsible for qualification designated in 6.4, plainly identified by securely attached durable tags marked with the following information:

Samples for Qualification tests
THERMOCOUPLES, CONTACT, AIRCRAFT ENGINE SPARK PLUG GASKET-TYPE
AN part number
Manufacturer's part number
Name of manufacturer
Submitted by (name) (date) for Qualification tests in accordance with the requirements of Specification MIL-T-5494A under authorization (reference authorizing letter).

4.3.2 Tests.- The Qualification tests of thermocouples shall consist of all the tests specified under Acceptance tests and, in addition, the following tests.

4.3.2.1 Full-range EMF output.- The EMF produced by the thermocouple shall be checked at the temperatures specified in table I and shall be as specified within a tolerance of ±0.16 mv.

4.3.2.2 Extreme temperature exposure.- The resistance of the thermocouple assembly shall be measured. The thermocouple assembly shall then be placed in an ambient temperature of -65°C (-85°F) for 48 hours, at the end of which time it shall be removed to an ambient temperature of 175°C (347°F) for 8 hours. The resistance of the thermocouple shall then be checked at room temperature. The change in resistance shall not exceed 0.005 ohm.

4.3.2.3 Vibration.- The thermocouple assembly shall be mounted on a vibration stand with the terminals fastened to a stationary panel in such a manner that the lead is parallel to the plane of the gasket and the radius of curvature of the lead, near the support, is approximately 1-1/2 inches. The vibration stand shall be such that a point on the thermocouple shall describe, in a plane inclined 45 degrees to the horizontal, a circle of diameter not less than 0.062 inch nor more than 0.100 inch, at any desired frequency between 500 and 3,000 cpm. The test shall be conducted with the thermocouple junction at a temperature of 250°C (482°F). There shall be no failure of any kind following 100 hours of vibration.

4.3.2.3.1 The EMF output test and Resistance test described in 4.4.2.1 and 4.4.1.2, respectively, shall be repeated following the vibration, and the thermocouple shall satisfactorily meet the requirements of these tests.

4.3.2.4 Humidity.- The thermocouple shall be placed in a suitable enclosure maintained at a temperature of 60° ±5°C (140° ±9°F) and relative humidity of 90 percent ±5 percent. After a period of 6 hours the heat shall be shut off and the enclosure allowed to cool for 18 hours. The humidity shall reach 100 percent as the temperature is reduced. (Precaution shall be taken to prevent water dripping on the thermocouples.) The entire procedure shall be repeated five times. The thermocouple shall then be subjected to the Resistance test specified in 4.4.1.2 and shall satisfy the requirements of that test. At the end of this test the EMF output test shall be repeated, and the thermocouple shall meet the requirements of that test.

4.3.2.5 Bending.- The Bending test as specified in 4.4.2.2 shall be repeated, except that the thermocouple shall be bent a total of two times.

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4.4 Acceptance tests.- The Acceptance tests shall consist of Individual tests and Sampling tests.

4.4.1 Individual tests.- Each thermocouple submitted for acceptance under contract shall be subjected to the following tests. In addition, thermocouples shall be subjected to any other test specified herein which the Inspector considers necessary to determine conformance with the requirements of this specification.

4.4.1.1 Examination of product.- Each thermocouple shall be examined for conformance with this specification with respect to material, workmanship, and marking.

4.4.1.2 Resistance.- The resistance of the complete thermocouple assembly shall be checked by any method satisfactory to the procuring activity and shall not exceed 0.15 ohm.

4.4.2 Sampling tests.- At least five thermocouples, selected by the Inspector at random from each 500 thermocouples, or less, of each lot, which have passed the required Individual tests, shall be subjected to the following tests. A lot shall consist of thermocouples of the same part number submitted for inspection under the same contract or order.

4.4.2.1 EMF output.- The EMF produced by the thermocouple shall be checked at a temperature of 200°C (392°F) and 300°C (572°F). The EMF shall be within the tolerances specified in table I.

4.4.2.2 Bending.- The copper gasket shall be firmly held on a test block by a spark plug. The metal support, near junction, shall then be bent to the same plane as the copper gasket and returned to the original position for a total of two times. There shall be no failure of the support, thermocouple wire, or insulation. The resistance shall be checked, and the change from the original value shall not exceed 0.005 ohm.

4.5 Rejection and retest.- When any representative sample fails to meet the requirements of the Sampling tests, the lot represented shall be rejected. Any thermocouple failing to meet the requirements of the Individual tests shall be rejected and returned. Thermocouples which have been rejected may be replaced or repaired to correct the defects and resubmitted for all specified tests. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the original defects shall be furnished the Inspector. Units rejected after retest shall not be resubmitted without the specific approval of the procuring activity.

5. PREPARATION FOR DELIVERY

5.1 Application.- The requirements of section 5 apply only to direct purchases by or direct shipments to the Government.

5.2 Packaging and packing.- Thermocouples shall be preserved and packaged in accordance with group X of Specification MIL-P-6064, type I, class II container. For domestic shipment, fiberboard conforming to Specification LLL-B-631 or LLL-B-636 may be substituted for that conforming to Specification JAN-P-108.

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5.3 Marking of shipments.- Interior packages and exterior shipping containers shall be marked in accordance with Standard MIL-STD-129. The identification shall be composed of the following information listed in the order shown:

Stock No. or other identification number as specified
in the purchase document*
THERMOCOUPLES, CONTACT, AIRCRAFT ENGINE SPARK PLUG GASKET-TYPE
Specification MIL-T-5494A
AN part number
Manufacturer's part number
Quantity
Name of contractor
Name of manufacturer (if not the same as contractor)
Contract or order number

*NOTE: The contractor shall enter the Federal Stock No. specified in the purchase document or as furnished by the procuring activity. When the Federal Stock No. is not provided or available from the procuring activity, leave space therefor and enter the Stock No. or other identification when provided by the procuring activity.

6. NOTES

6.1 Intended use.- The thermocouples covered by this specification are intended for use on aircraft to actuate a thermocouple-type of indicator for indicating the cylinder head temperature of air-cooled engine cylinders.

6.2 Ordering data.- Procurement documents should specify the title, number, and date of this specification, and the size thermocouple desired.

6.3 Superseding data.- This specification supersedes Specification MIL-T-5494, which specification incorporated the requirements of Specification AN-T-76.

6.4 Provisions for qualification.- With respect to products requiring qualification, awards will be made only for such products as have, prior to the bid opening date, been tested and approved for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date.

6.4.1 The attention of suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government, tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Requests for information pertaining to qualification of products covered by this specification should be addressed to the Commander, Wright Air Development Center, Wright-Patterson Air Force Base, Ohio, the activity responsible for qualification, with a copy to the Bureau of Aeronautics, Navy Department, Washington 25, D. C.

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